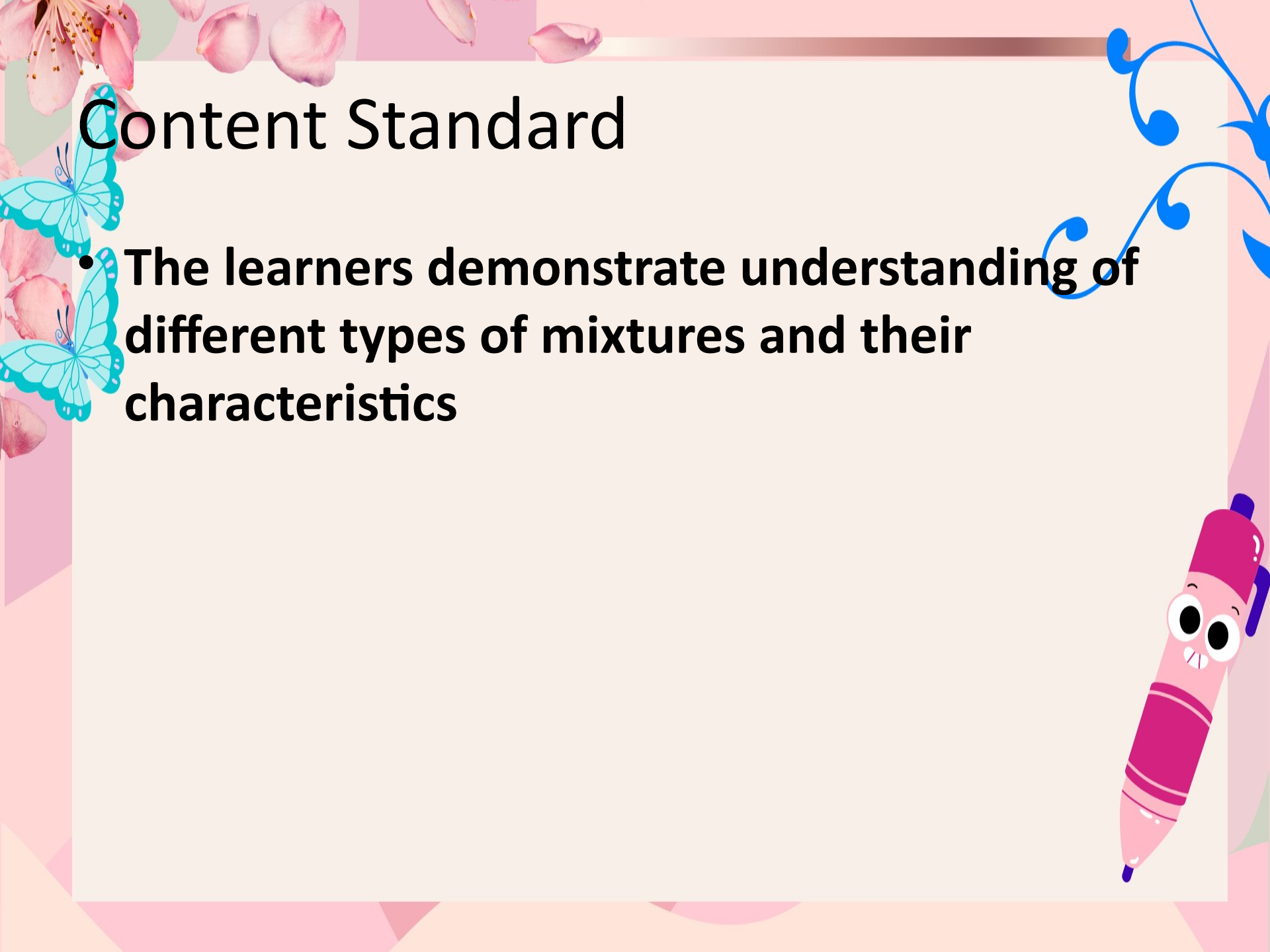


SCIENCE 6

(Describing Mixtures)

Quarter 1
Week 1 – Day 1





Content Standard

- **The learners demonstrate understanding of different types of mixtures and their characteristics**



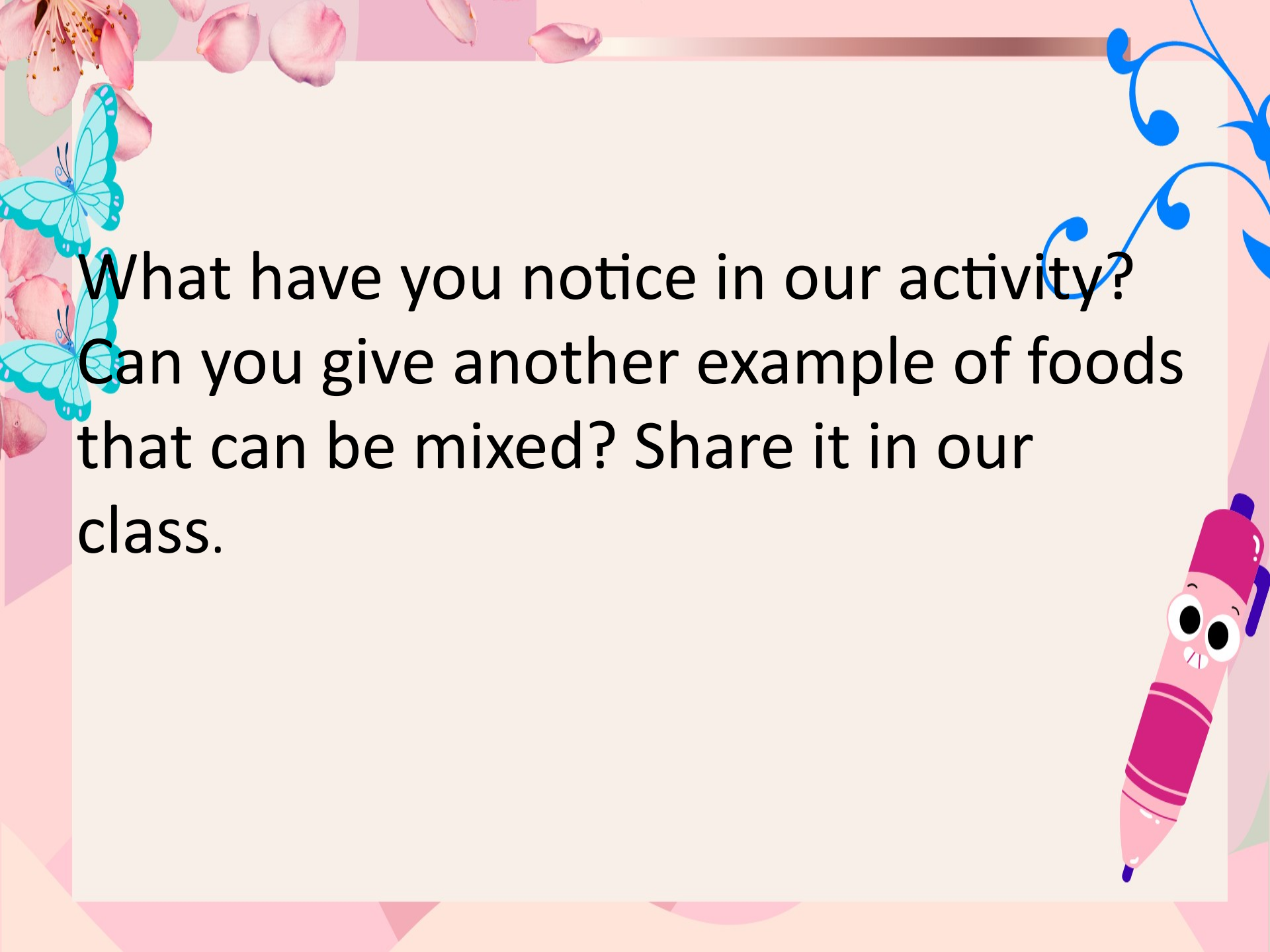
Motivation:

Have you tried to make a sandwich? How can we make a sandwich? Can you arrange the ingredients below?

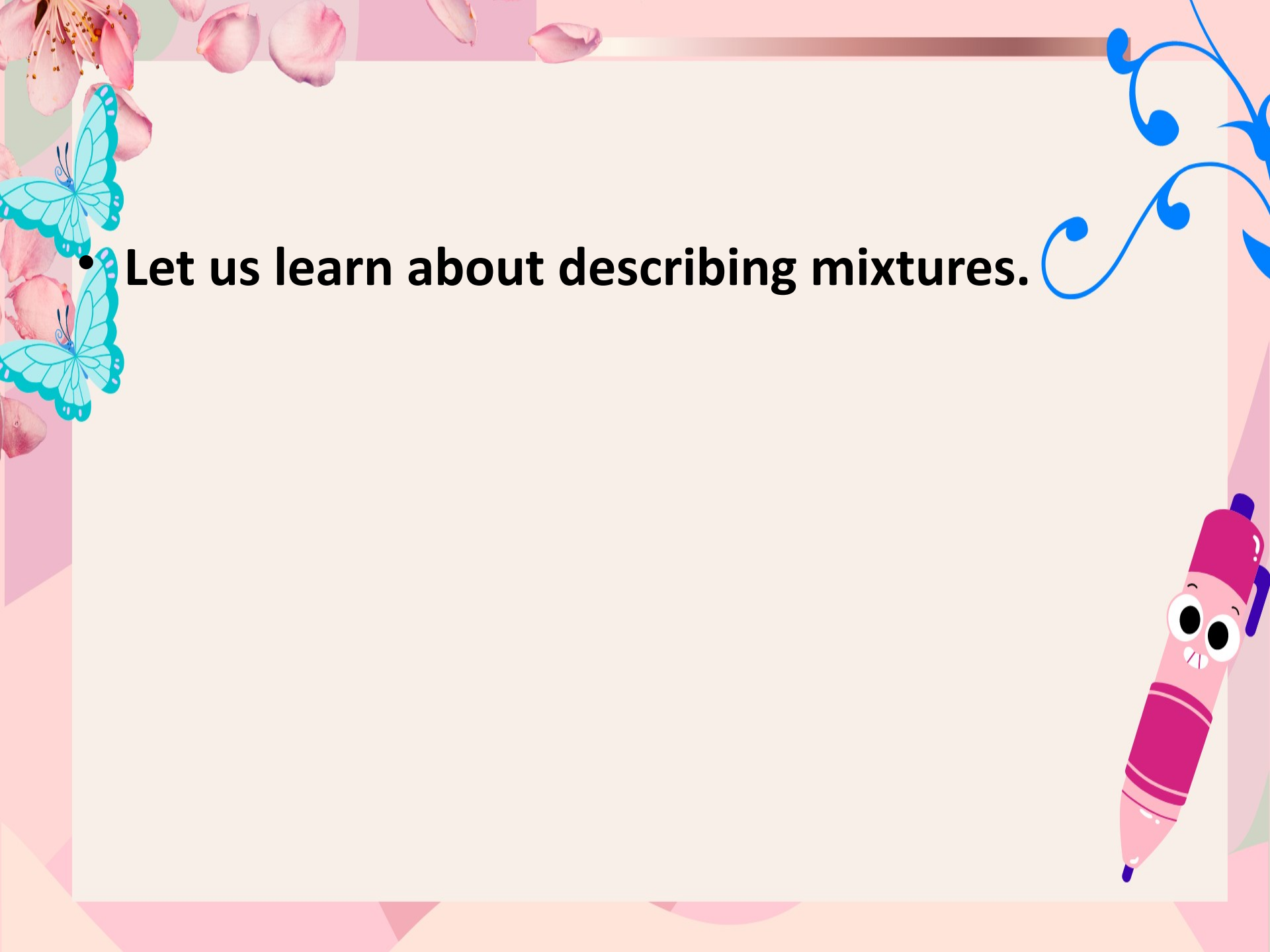
How to make sandwich?

_____ + _____ + _____ = _____





What have you notice in our activity?
Can you give another example of foods
that can be mixed? Share it in our
class.



• **Let us learn about describing mixtures.**



Discussion

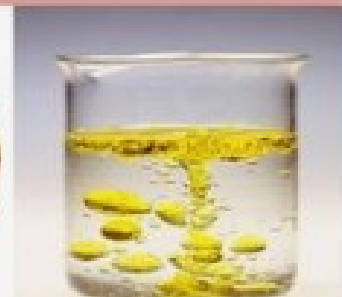
- Mixtures are combinations of two or more substances that can be homogeneous or heterogeneous.

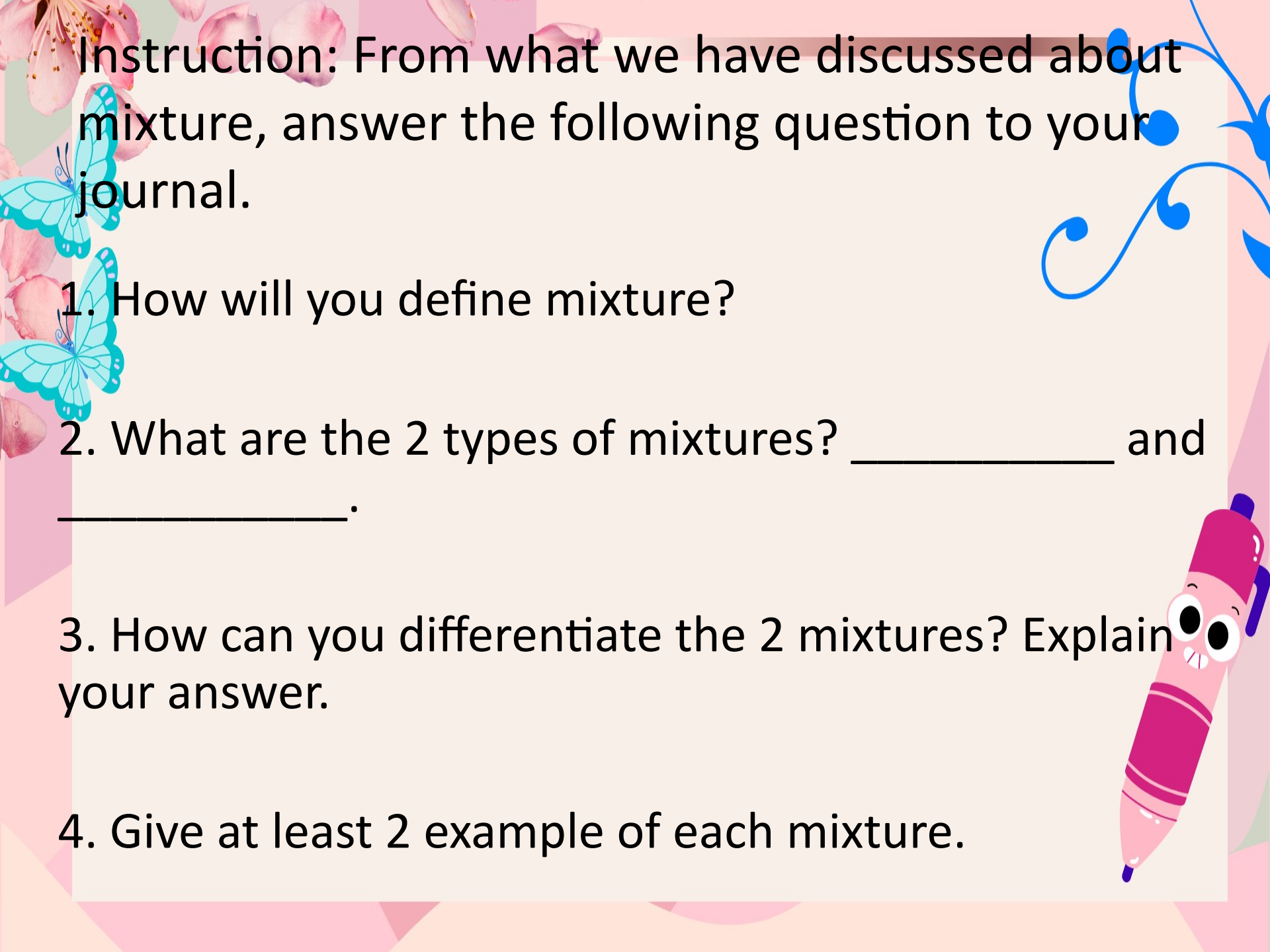
Homogeneous mixtures appear uniform all throughout because they have the same proportion. Heterogeneous mixtures are not uniform in proportion where combined substances are not evenly spread or mixed.



Types of Mixtures

- Heterogeneous mixtures
 - Can see the different parts of the mixture
 - Liquids are unstable and will settle out
 - EX. Pizza, Oil & water, etc.
- Homogeneous mixtures
 - Can NOT see the different parts of the mixture
 - Solutions
 - Liquids are stable and typically won't settle
 - EX. Alloys (metal mixtures), Salt water, etc.





Instruction: From what we have discussed about mixture, answer the following question to your journal.

1. How will you define mixture?

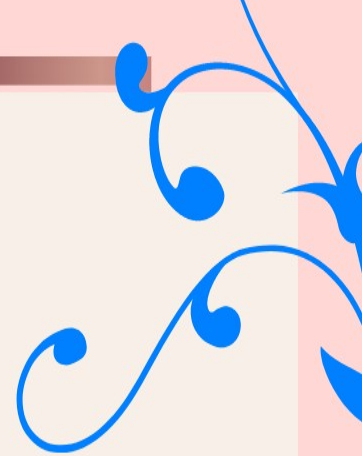
2. What are the 2 types of mixtures? _____ and _____.

3. How can you differentiate the 2 mixtures? Explain your answer.

4. Give at least 2 example of each mixture.



- **Discuss the concept of the lesson.**



Instruction: Identify if the following mixture is homogenous or heterogeneous.

1.



2.



3

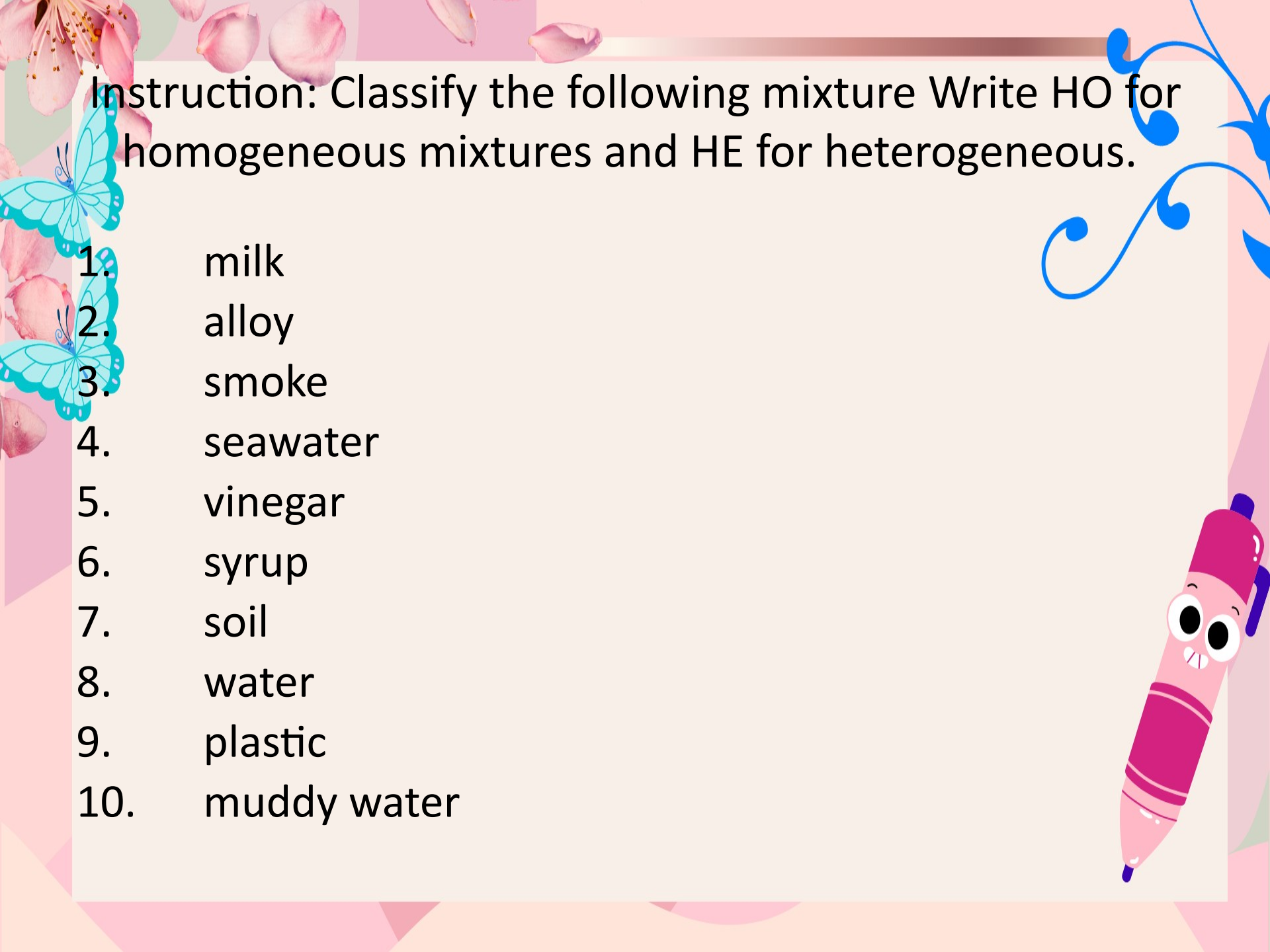


4.



5.





Instruction: Classify the following mixture Write HO for homogeneous mixtures and HE for heterogeneous.

1. milk
2. alloy
3. smoke
4. seawater
5. vinegar
6. syrup
7. soil
8. water
9. plastic
10. muddy water

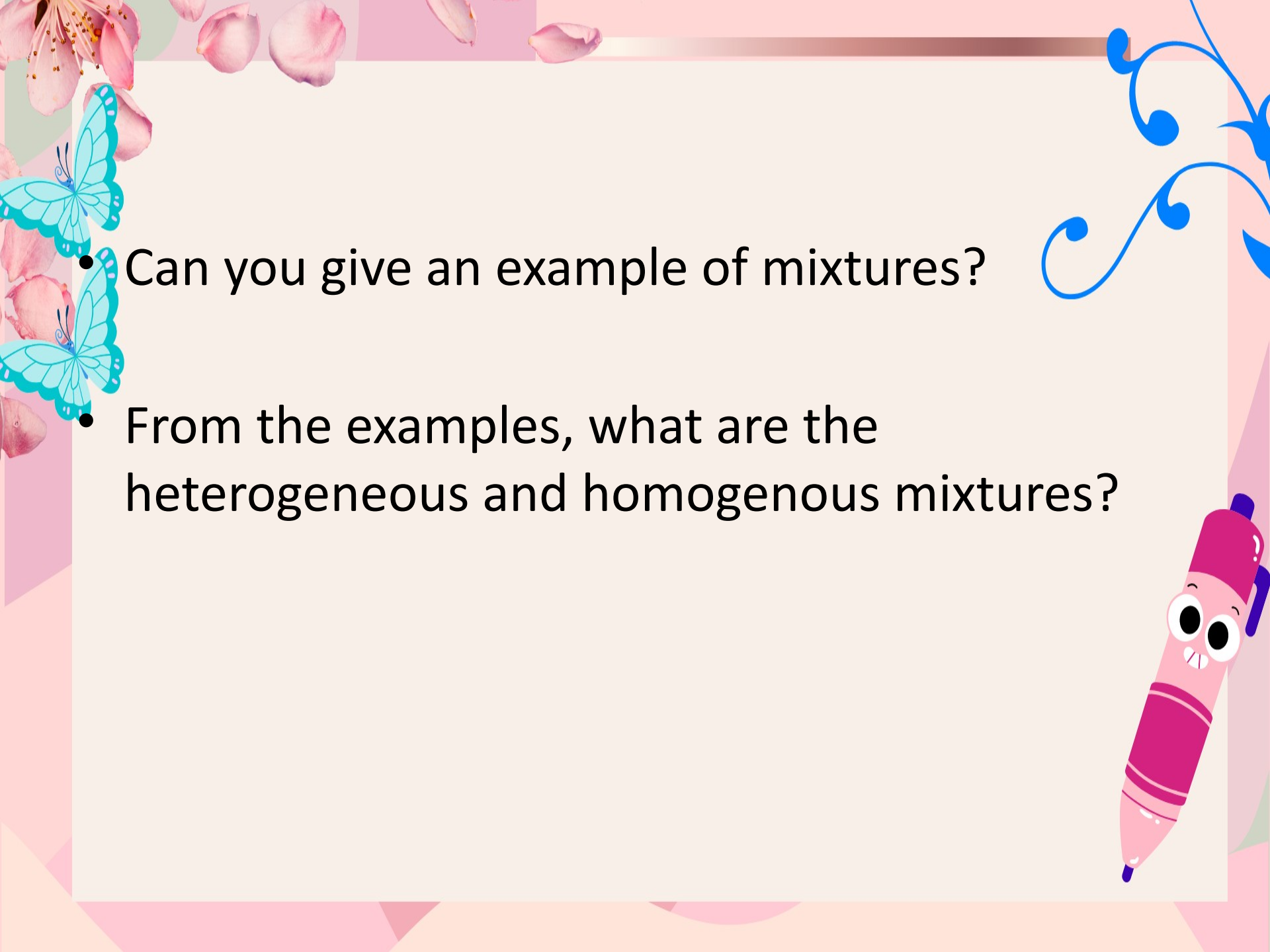


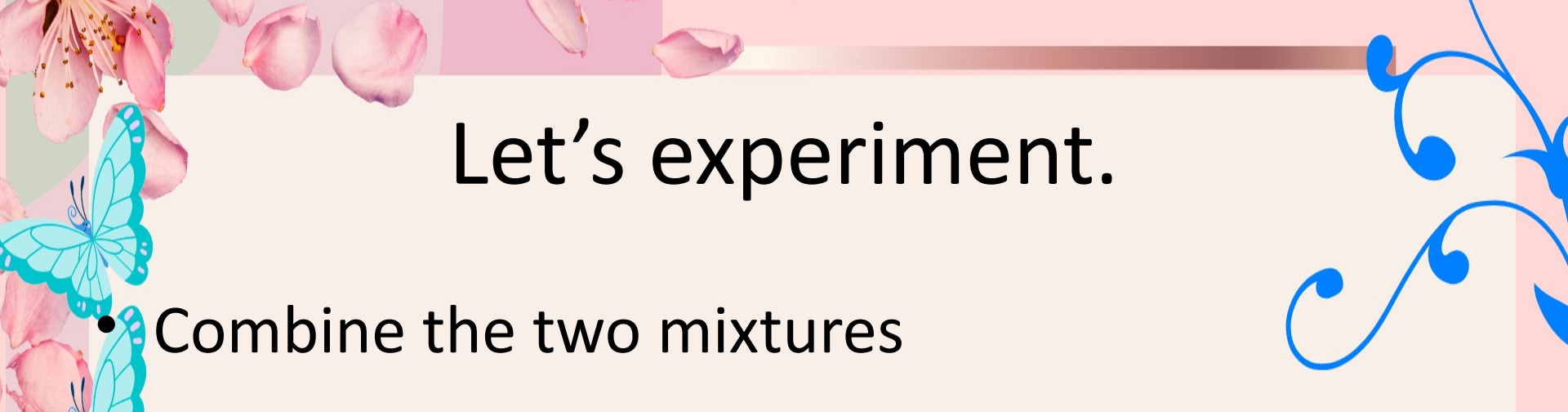
SCIENCE 6

(Describing Mixtures)

Quarter 1
Week 1 – Day 2



- 
- Can you give an example of mixtures?
 - From the examples, what are the heterogeneous and homogenous mixtures?

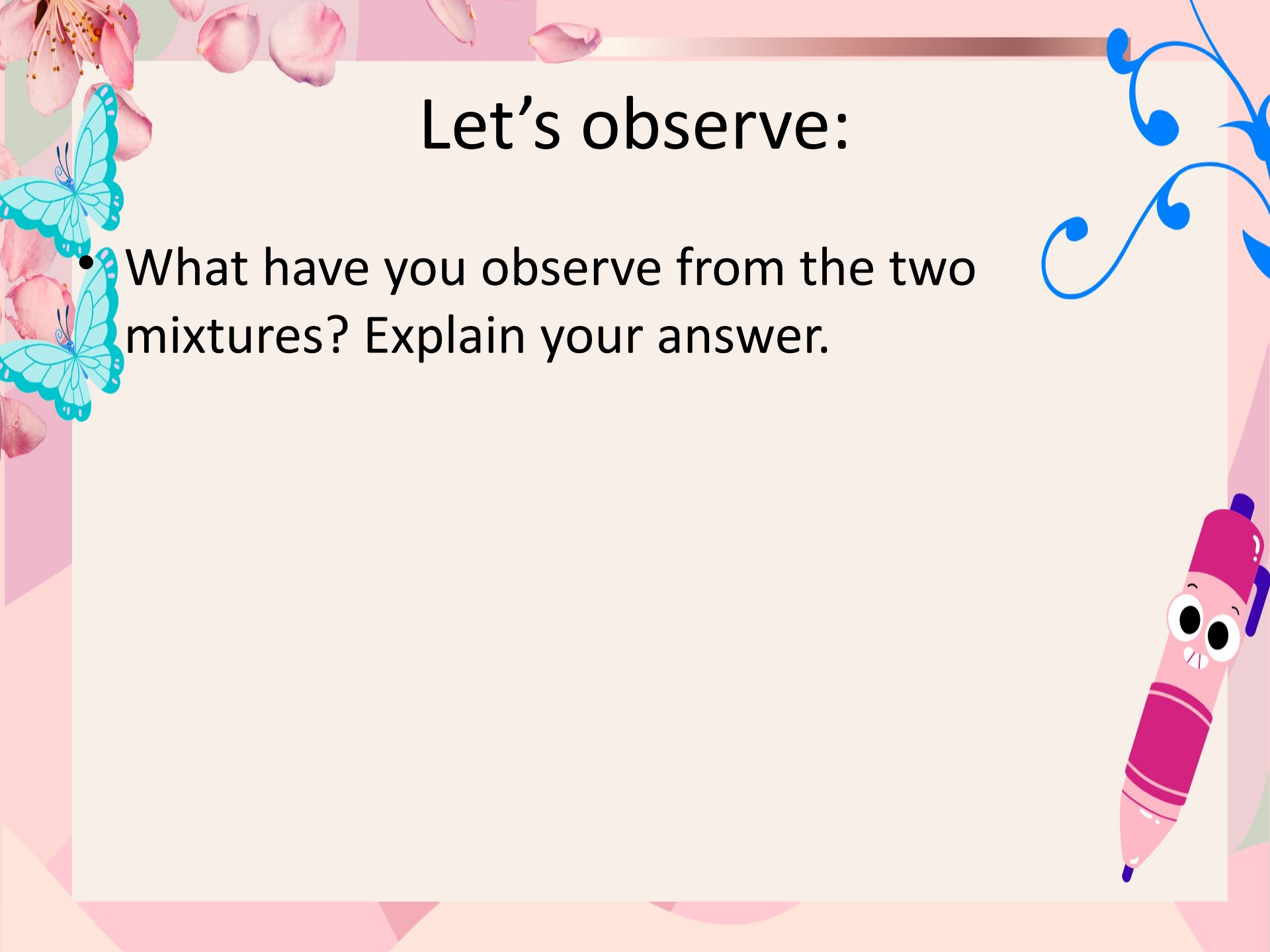


Let's experiment.

- Combine the two mixtures

1. oil and water
2. sugar and water





Let's observe:

- What have you observe from the two mixtures? Explain your answer.

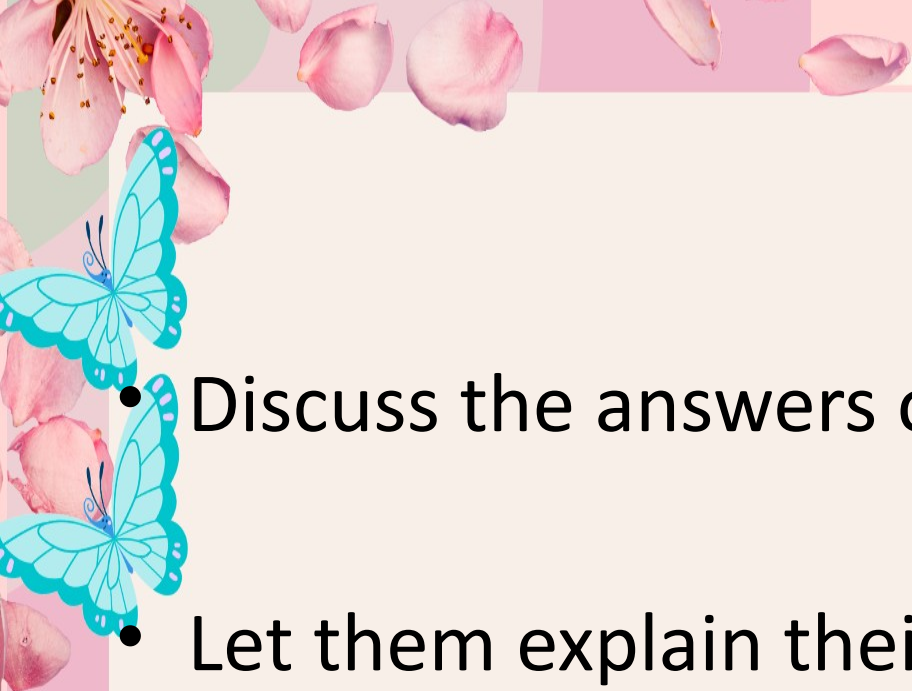
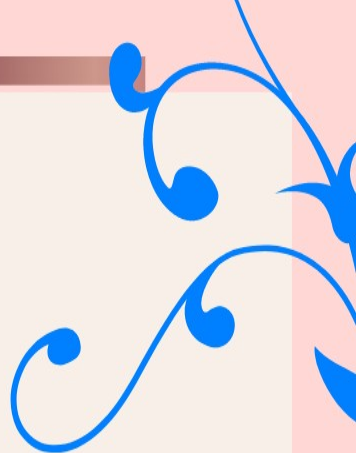


Let's do this.

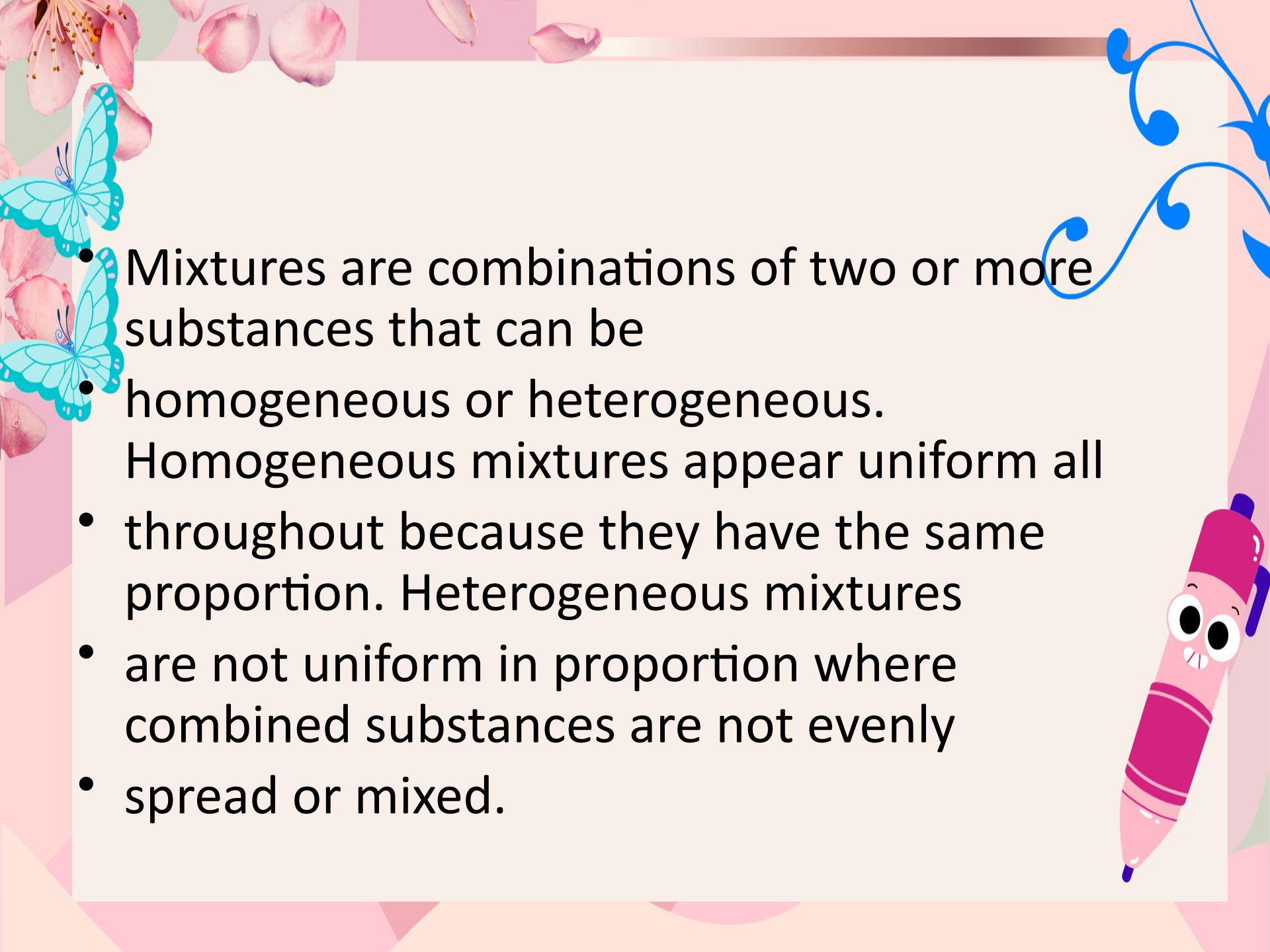
- Prepare solid and liquid materials that the learners will mixed.
- Ask the pupils to combine two substance and list the mixture on the board. Identify if it is heterogeneous and homogeneous.

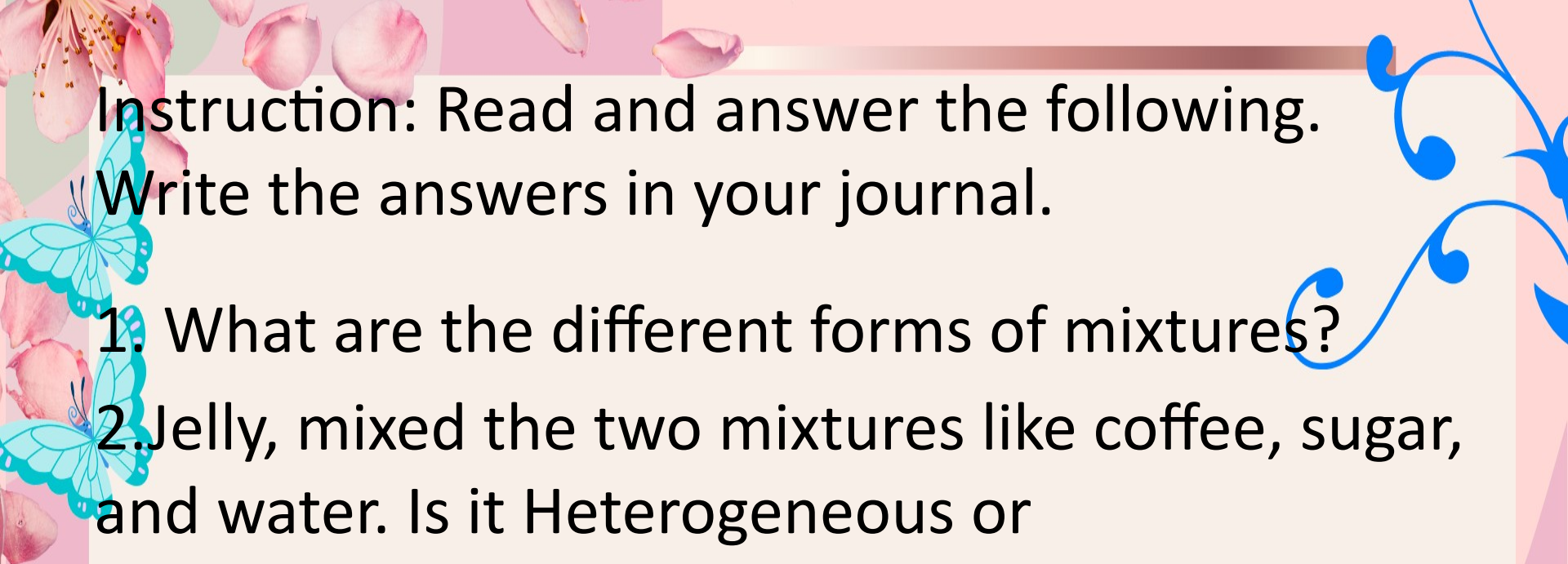
Mixture	Homogeneous	Heterogeneous



- 
- 
- Discuss the answers of the pupils.
 - Let them explain their answer for understanding.



- 
- Mixtures are combinations of two or more substances that can be
 - homogeneous or heterogeneous.
Homogeneous mixtures appear uniform all
 - throughout because they have the same proportion. Heterogeneous mixtures
 - are not uniform in proportion where combined substances are not evenly
 - spread or mixed.



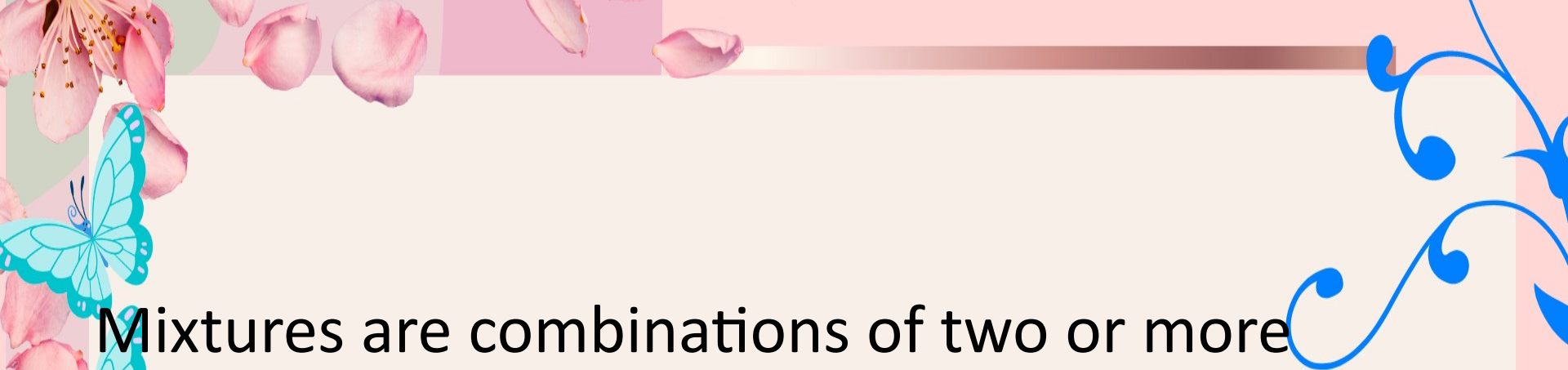
Instruction: Read and answer the following.
Write the answers in your journal.

1. What are the different forms of mixtures?

2. Jelly, mixed the two mixtures like coffee, sugar, and water. Is it Heterogeneous or Homogeneous? Why?

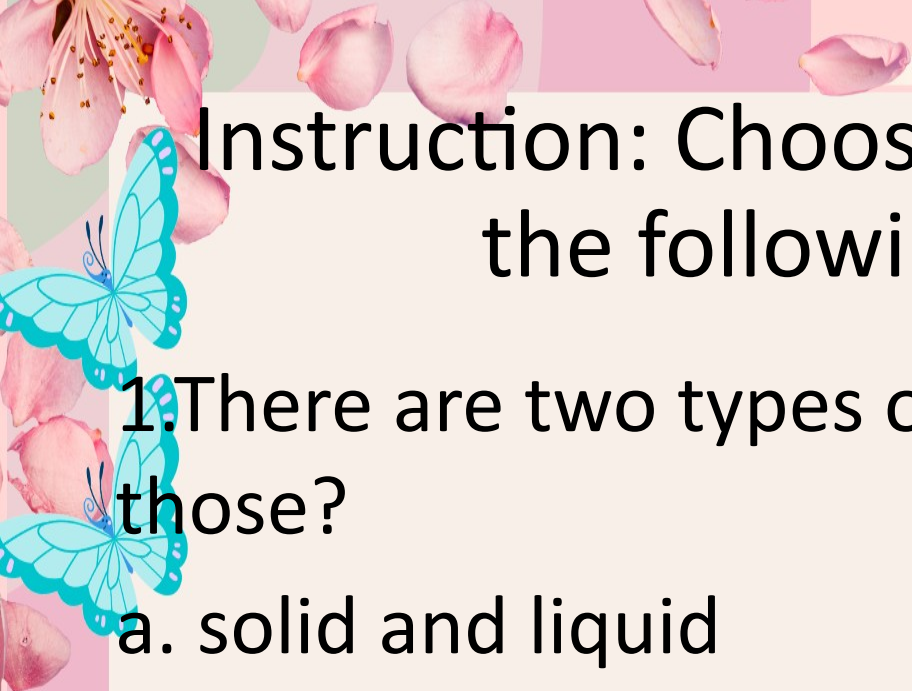
3. Shiela, combined the different vegetables to make a salad. Is it Heterogeneous or Homogeneous? Why?





Mixtures are combinations of two or more substances that can be homogeneous or heterogeneous.



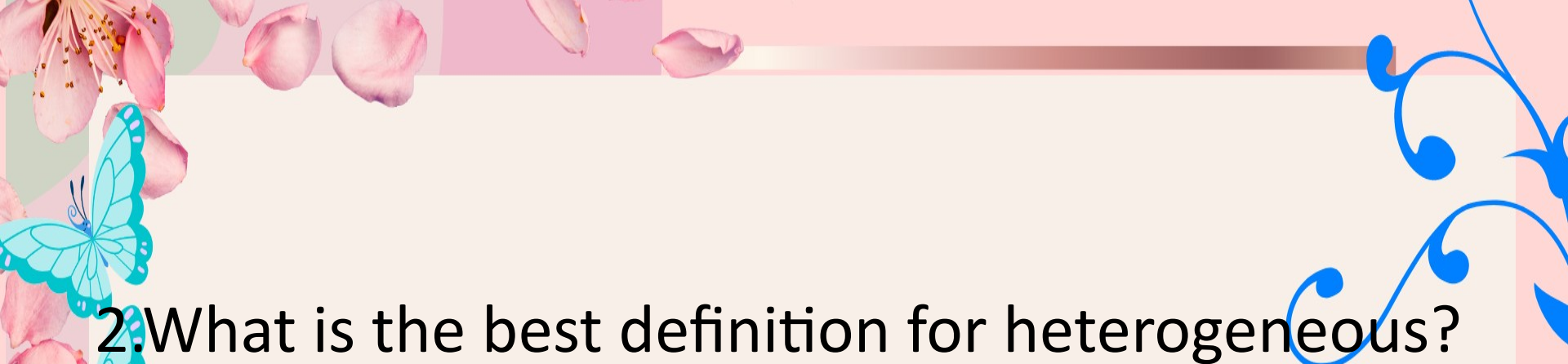


Instruction: Choose the best answer in the following questions.

1. There are two types of mixture. What are those?

- a. solid and liquid
- b. Liquid and gas
- c. heterogeneous and gas substance
- d. heterogeneous and homogeneous

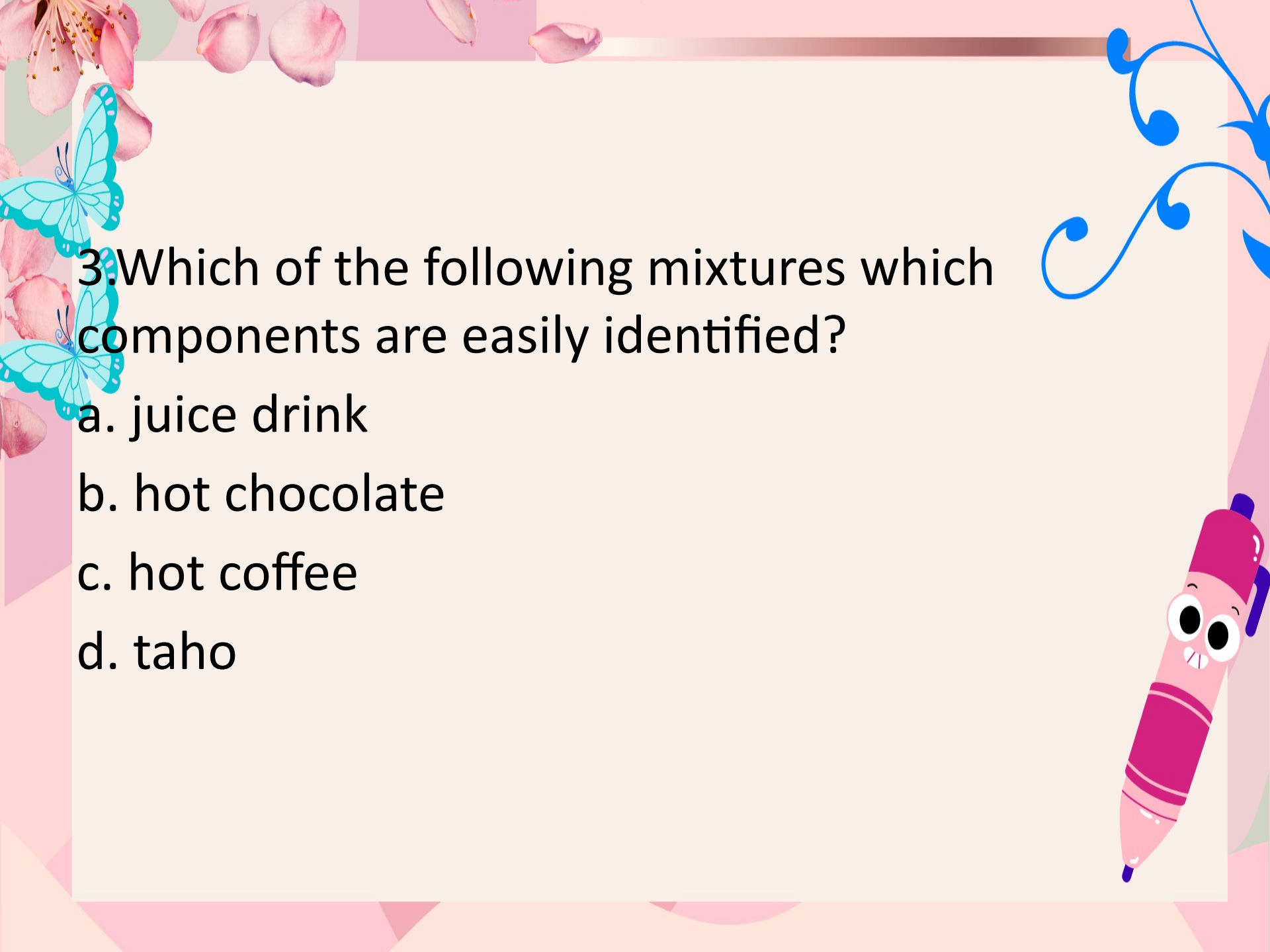




2. What is the best definition for heterogeneous?

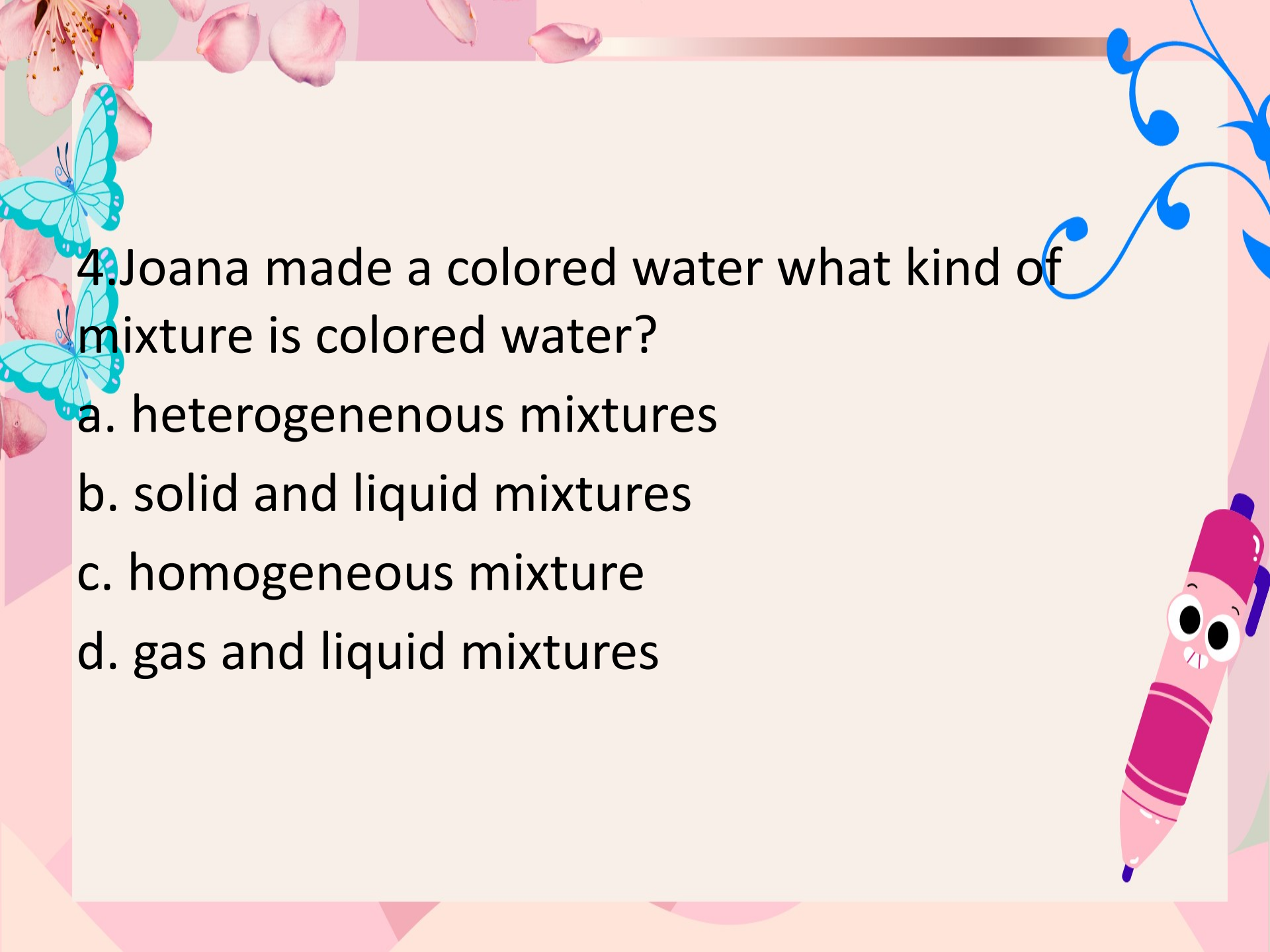
- a. We cannot distinguish their components.
- b. It is made of solid and liquid substances.
- c. It is made of liquid and gas substances.
- d. we can distinguish their components.





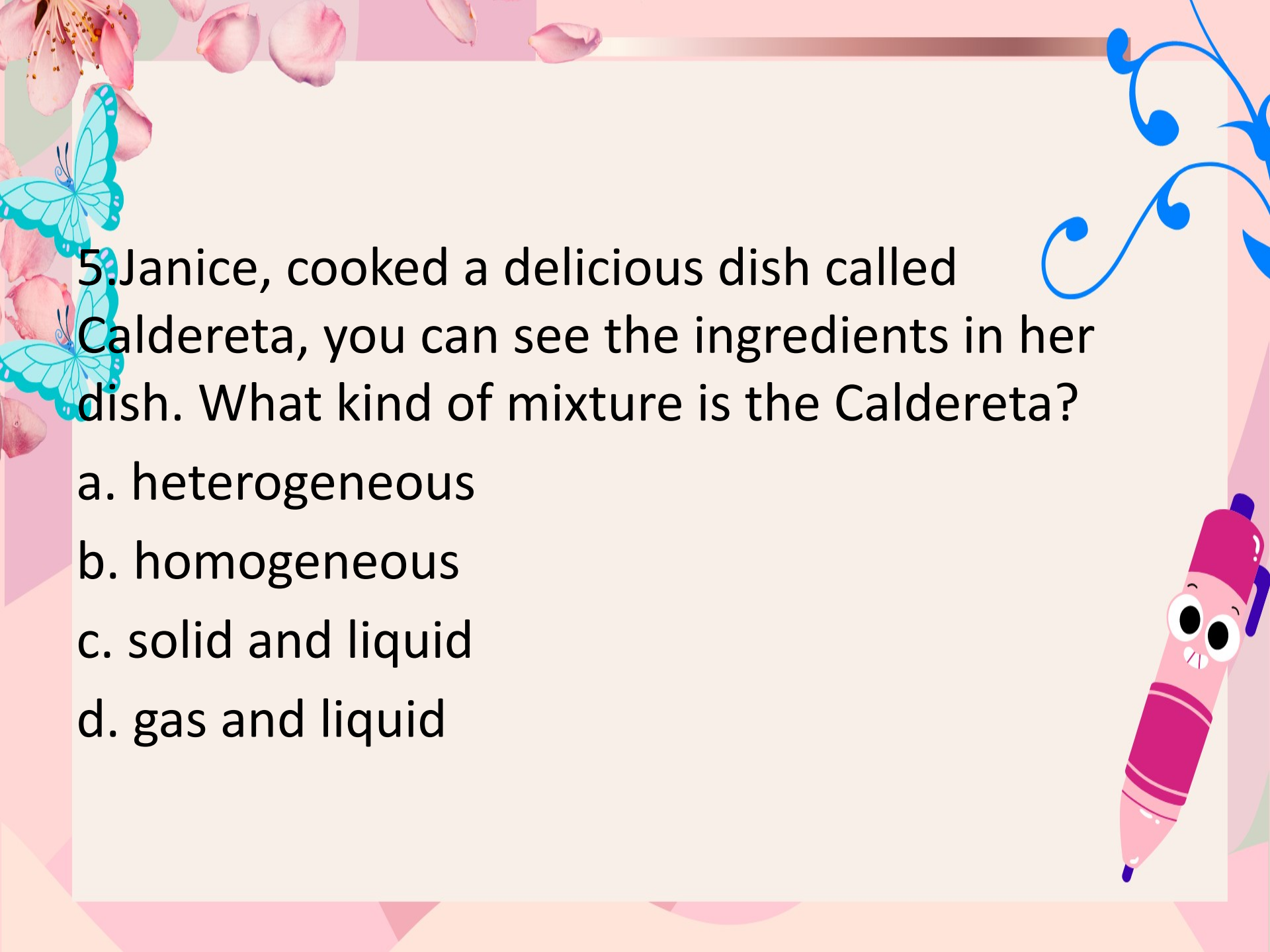
3. Which of the following mixtures which components are easily identified?

- a. juice drink
- b. hot chocolate
- c. hot coffee
- d. taho



4. Joana made a colored water what kind of mixture is colored water?

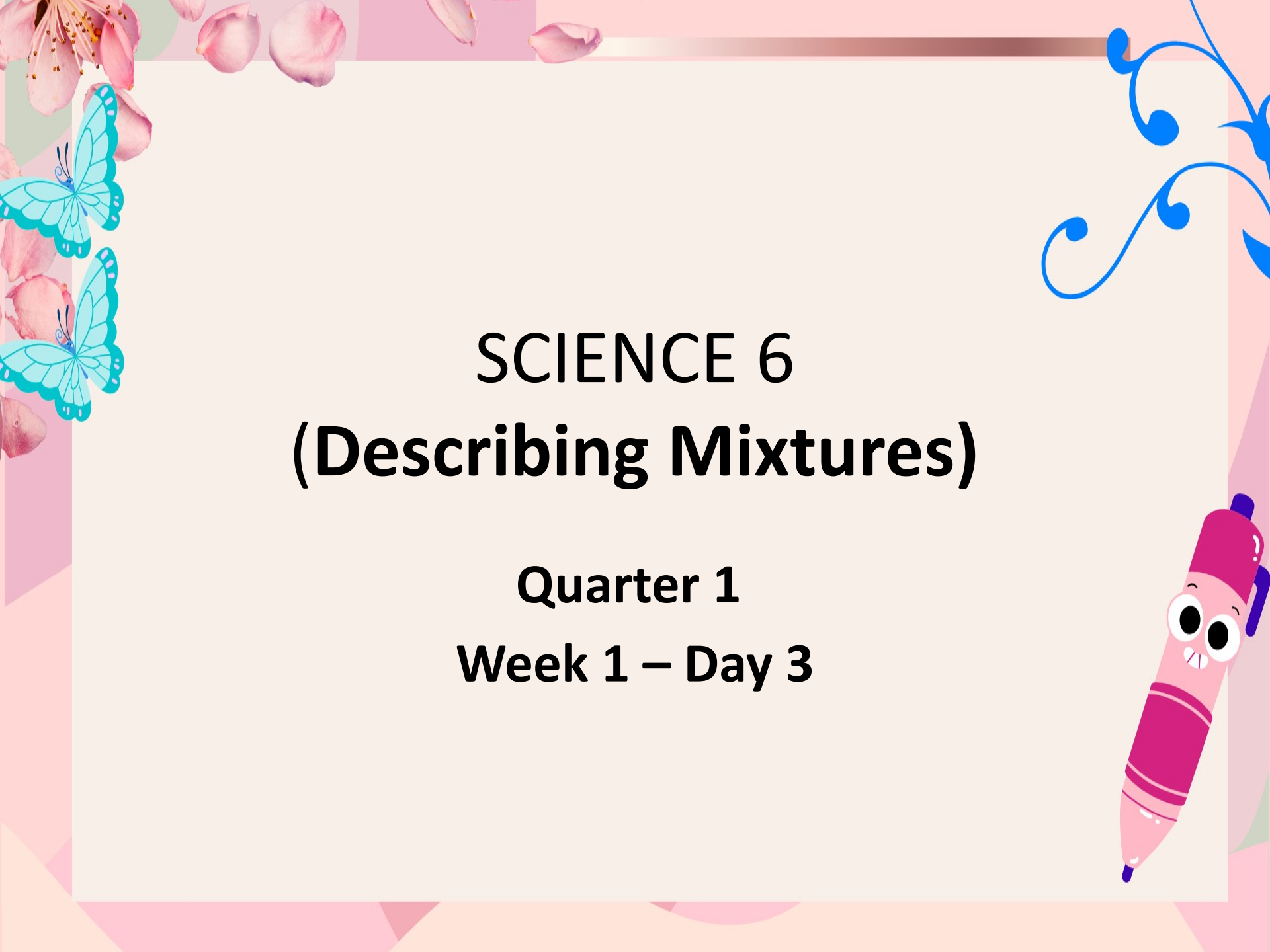
- a. heterogenous mixtures
- b. solid and liquid mixtures
- c. homogeneous mixture
- d. gas and liquid mixtures



5. Janice, cooked a delicious dish called Caldereta, you can see the ingredients in her dish. What kind of mixture is the Caldereta?

- a. heterogeneous
- b. homogeneous
- c. solid and liquid
- d. gas and liquid

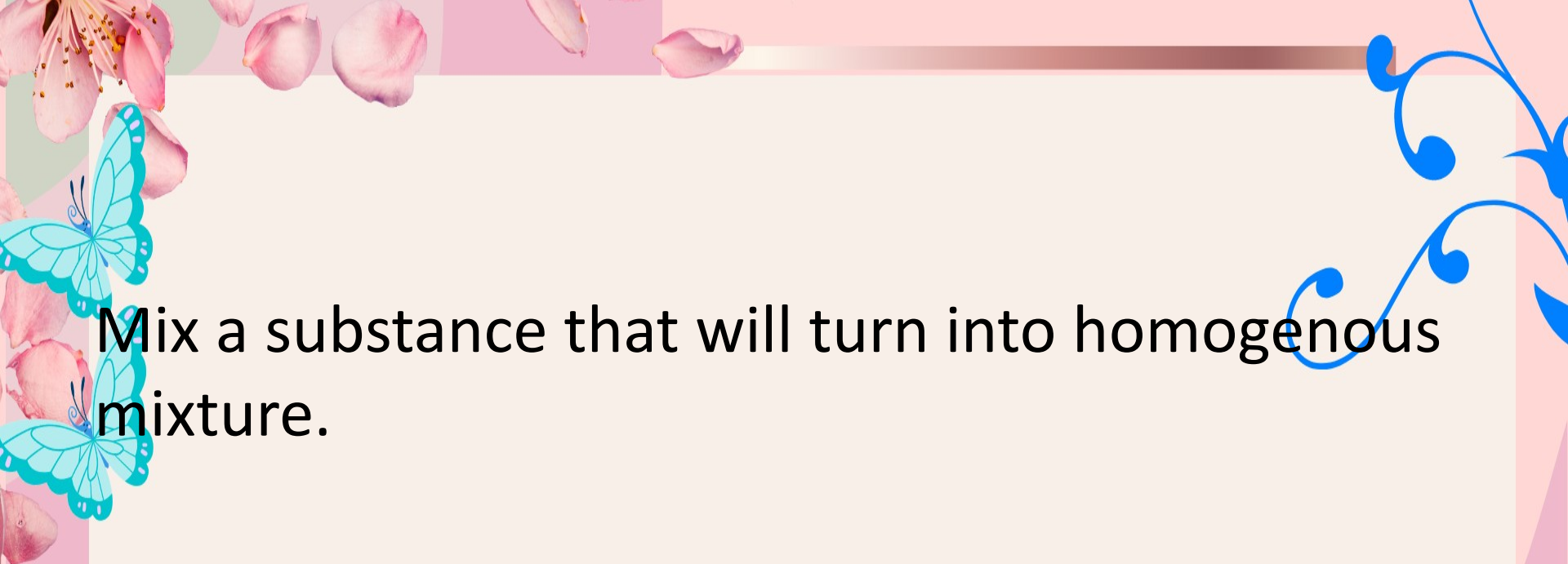




SCIENCE 6

(Describing Mixtures)

Quarter 1
Week 1 – Day 3



Mix a substance that will turn into homogenous mixture.

1. ___ + ___ = ___

2. ___ + ___ = ___

3. ___ + ___ = ___

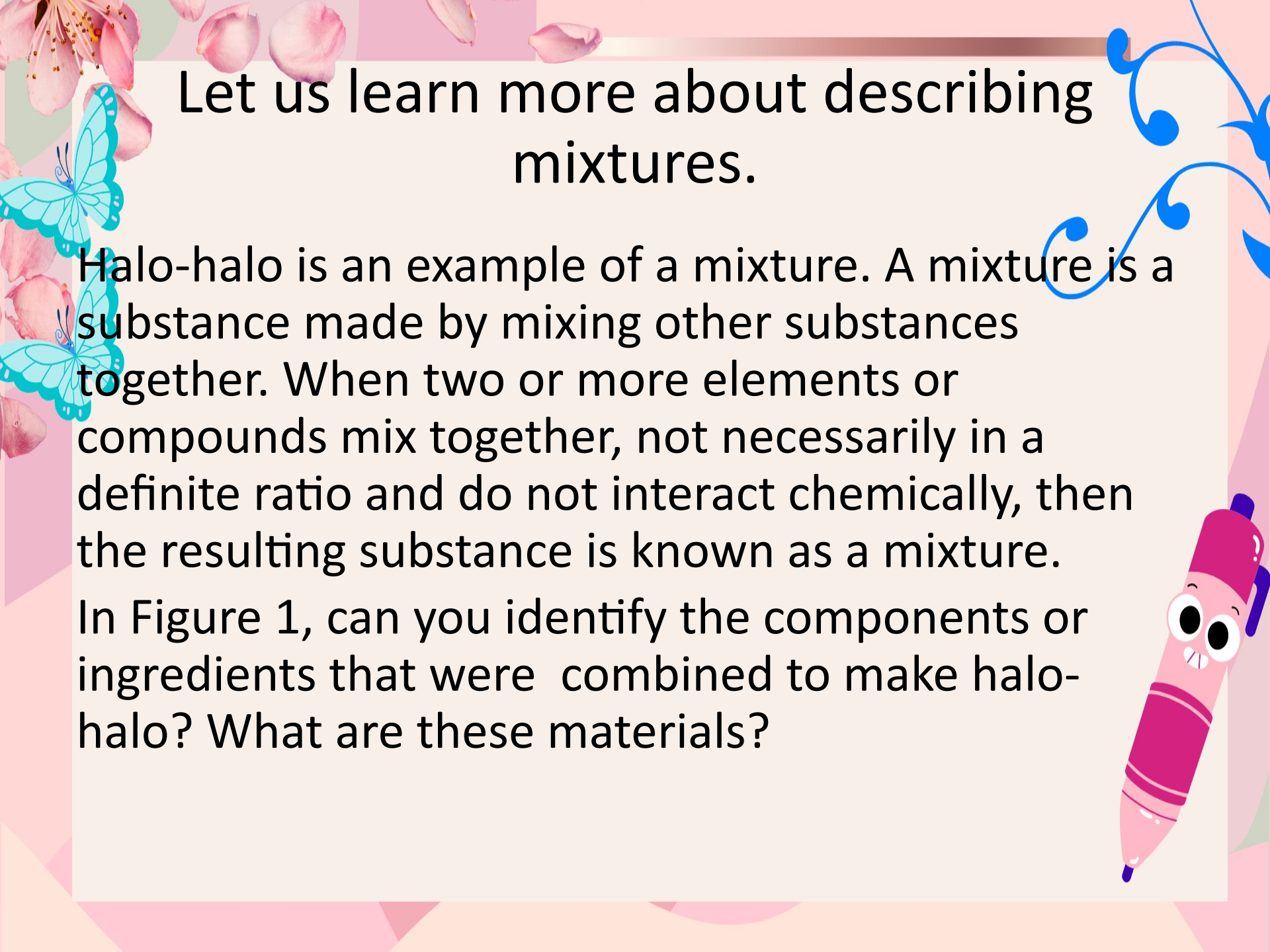
4. ___ + ___ = ___

5. ___ + ___ = ___



Look at the picture and describe it.



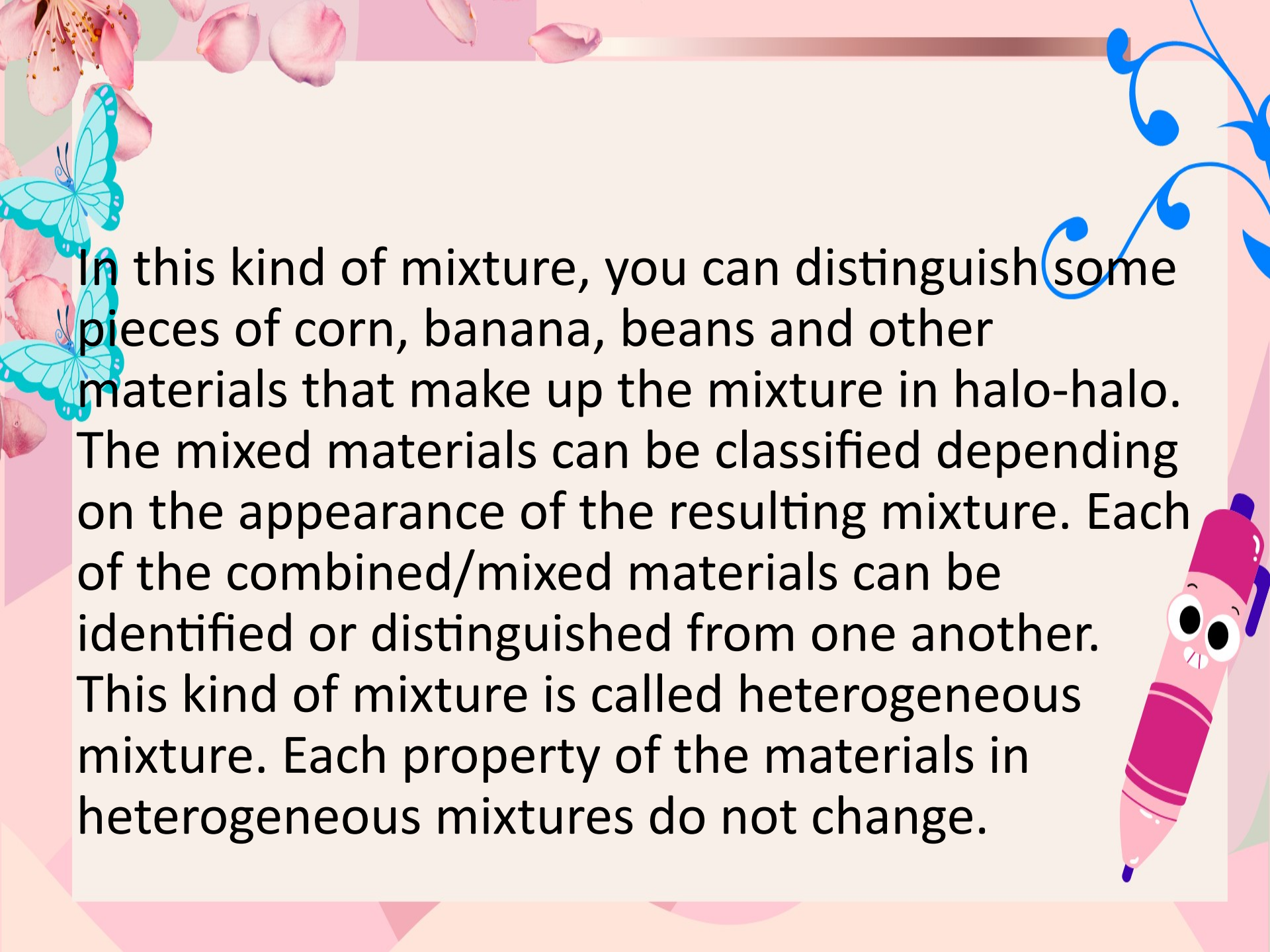


Let us learn more about describing mixtures.

Halo-halo is an example of a mixture. A mixture is a substance made by mixing other substances together. When two or more elements or compounds mix together, not necessarily in a definite ratio and do not interact chemically, then the resulting substance is known as a mixture.

In Figure 1, can you identify the components or ingredients that were combined to make halo-halo? What are these materials?

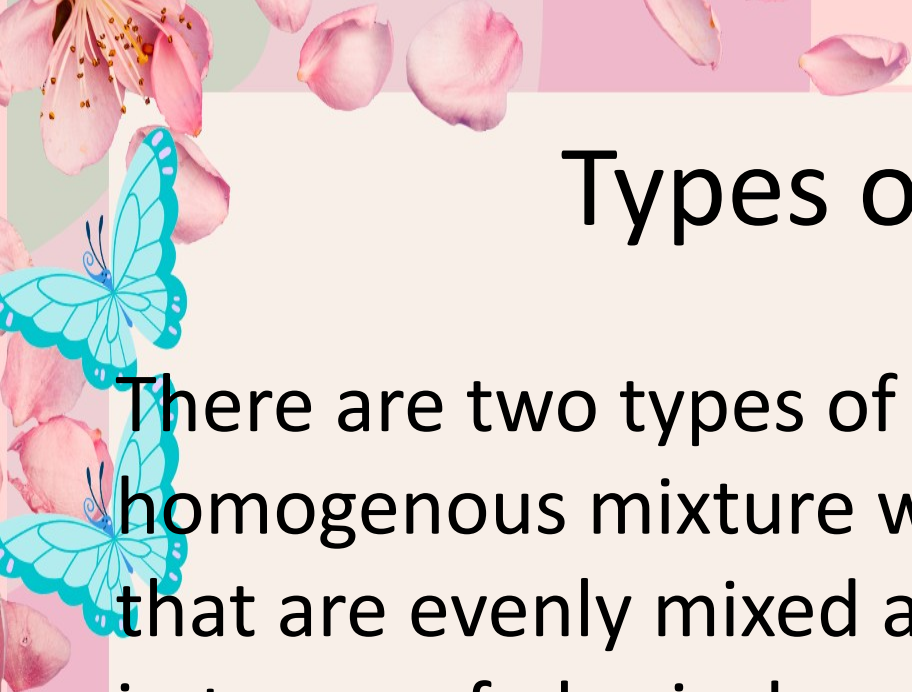
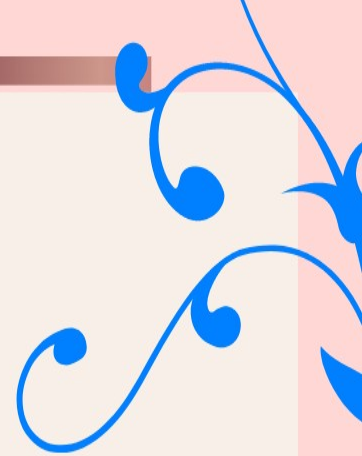




In this kind of mixture, you can distinguish some pieces of corn, banana, beans and other materials that make up the mixture in halo-halo. The mixed materials can be classified depending on the appearance of the resulting mixture. Each of the combined/mixed materials can be identified or distinguished from one another. This kind of mixture is called heterogeneous mixture. Each property of the materials in heterogeneous mixtures do not change.

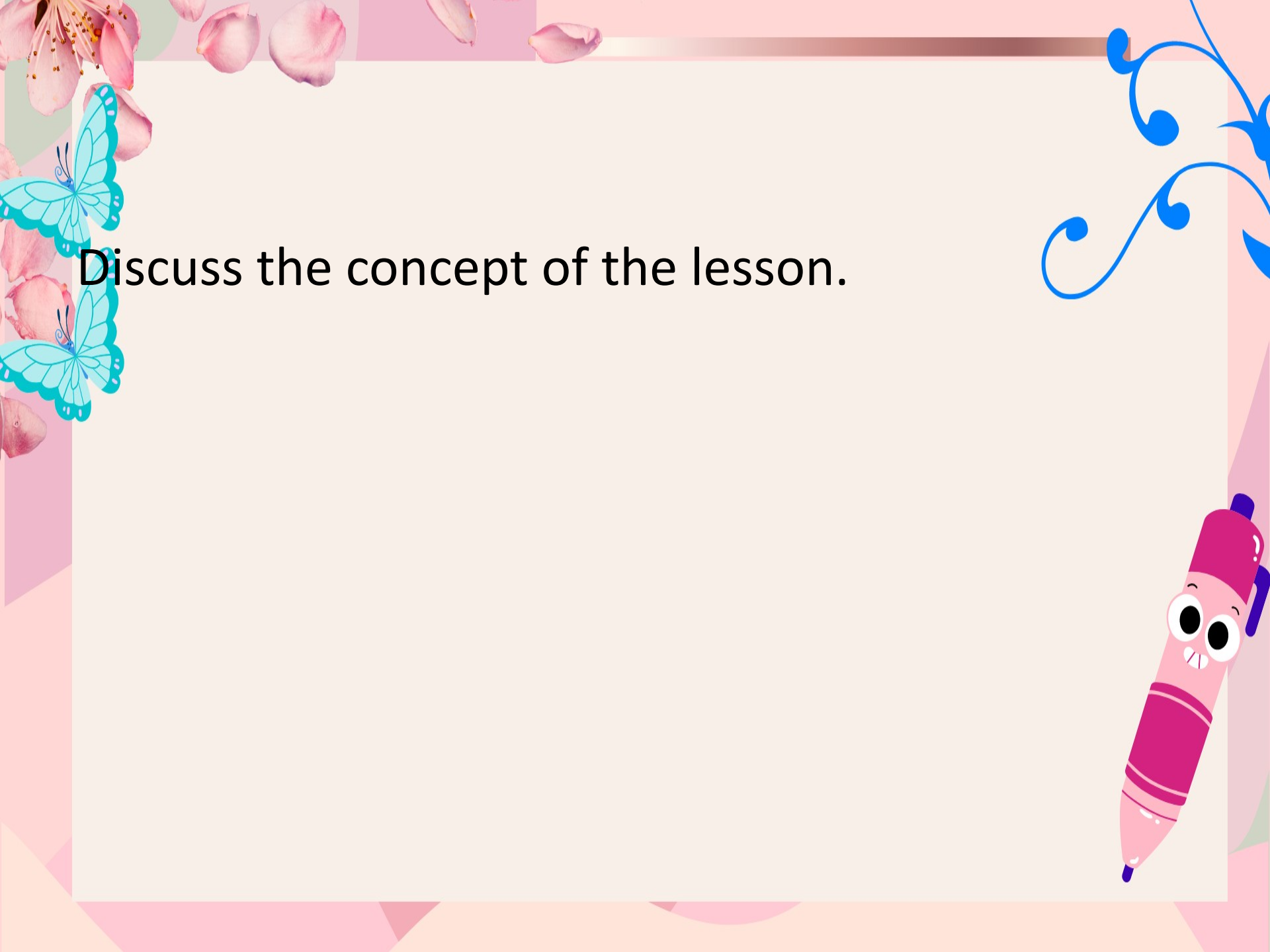
Types of Mixture

There are two types of mixtures. One is homogenous mixture which contains substances that are evenly mixed and is uniform throughout in terms of physical appearance. The other one is heterogeneous mixture which contains component substances that are usually easy to distinguish.



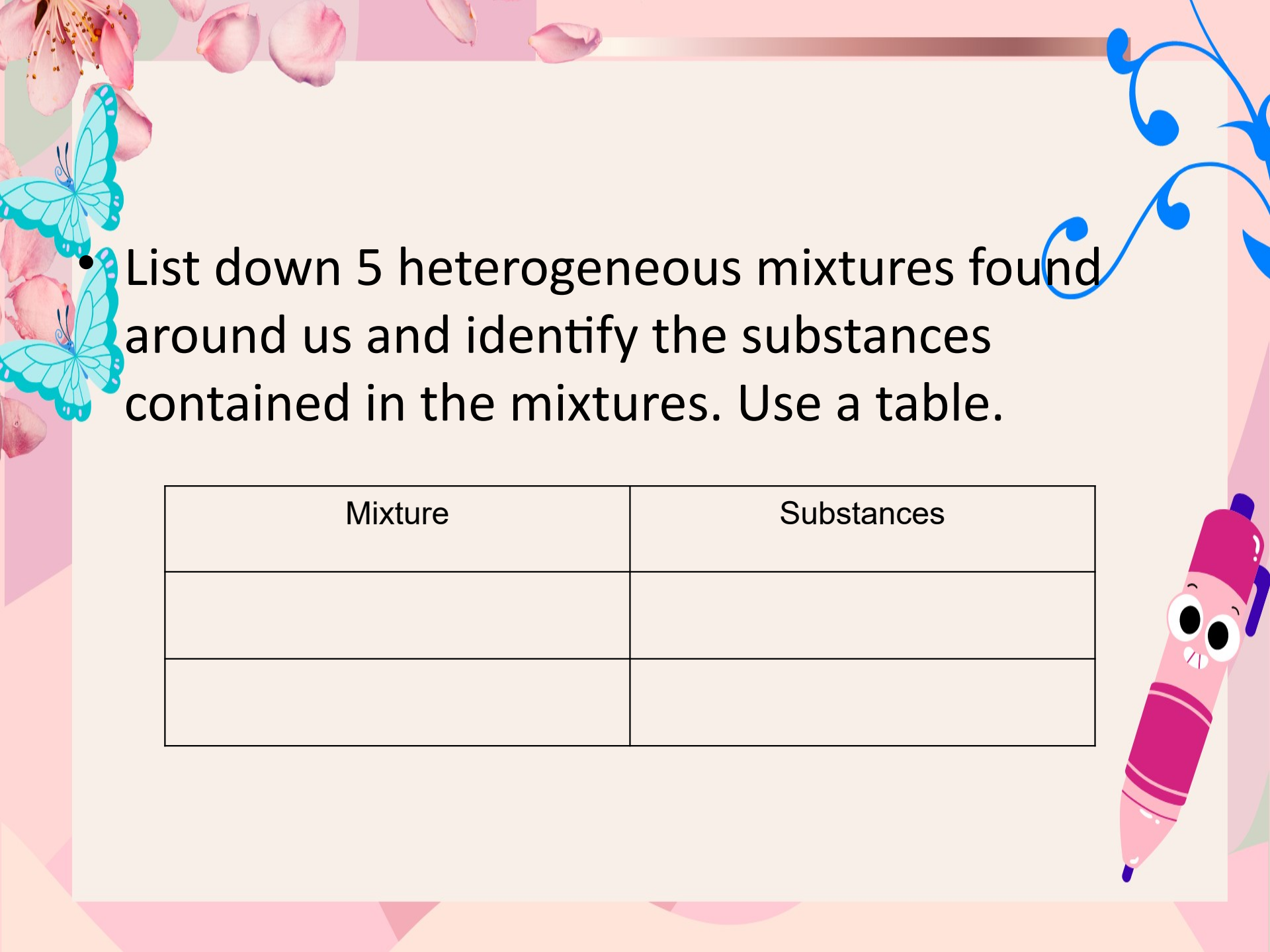
Examine the pictures in Figure 2. Can you identify the materials that make up these mixtures?





Discuss the concept of the lesson.



- 
- List down 5 heterogeneous mixtures found around us and identify the substances contained in the mixtures. Use a table.

Mixture	Substances



HOMOGENEOUS MIXTURES VERSUS HETEROGENEOUS MIXTURES

Homogenous mixtures have a uniform composition throughout the mixture

Components are not visible to the naked eye

The whole mixture is in the same phase

Particle size is often at atomic or molecular level

Components cannot be separated easily

Heterogenous mixtures have a mixed composition which may vary from point to point

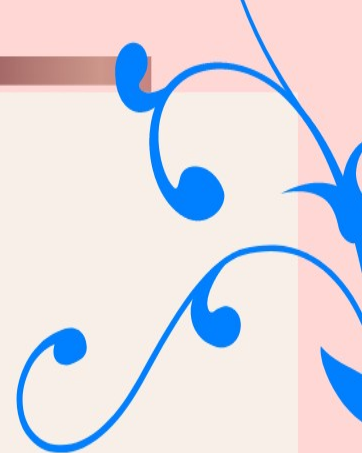
Components can be seen easily

Substances can be of two phases and layers may separate

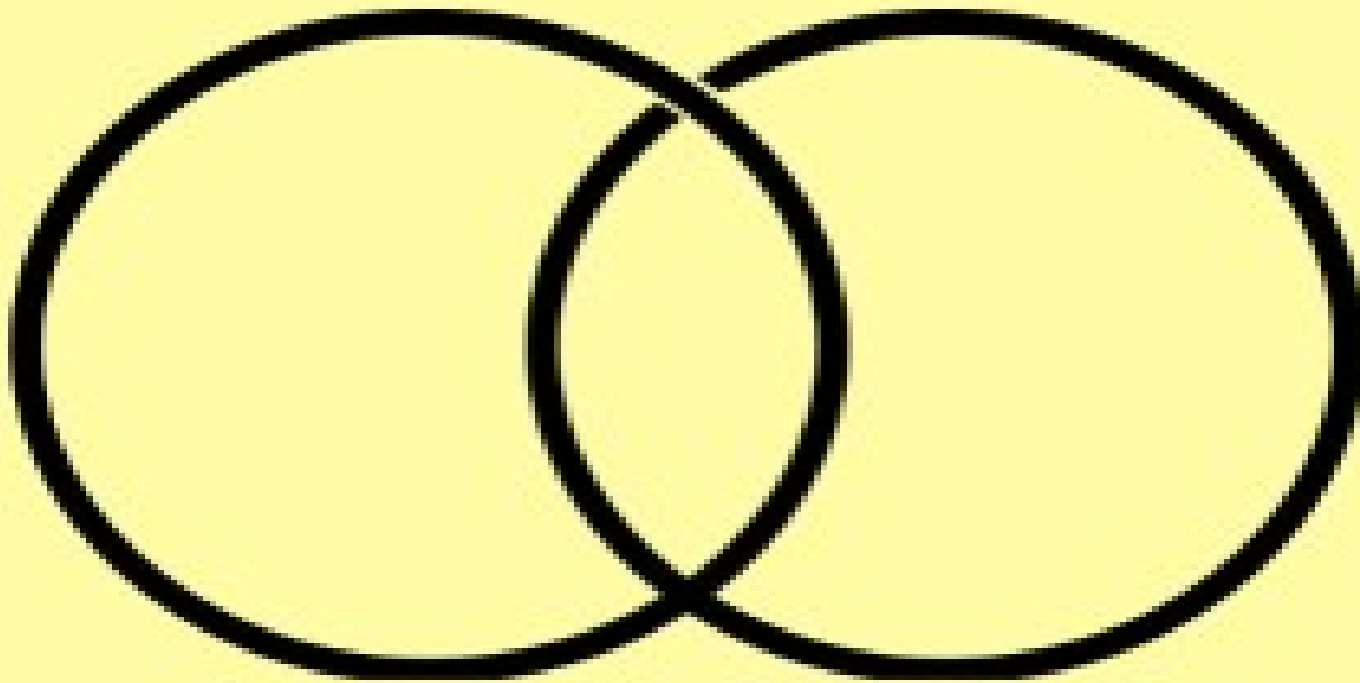
Heterogenous mixtures have large particle sizes

Components can be separated easily

Visit www.pediasa.com



Instruction: Complete the Venn diagram. Write the differences and similarities of two mixtures





SCIENCE 6

(Describing Mixtures)

Quarter 1
Week 1 – Day 4



Let's mix!

Mix a substance that will turn into heterogeneous mixture.

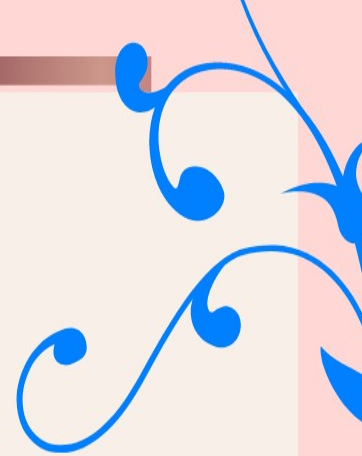
1. + =

2. + =

3. + =

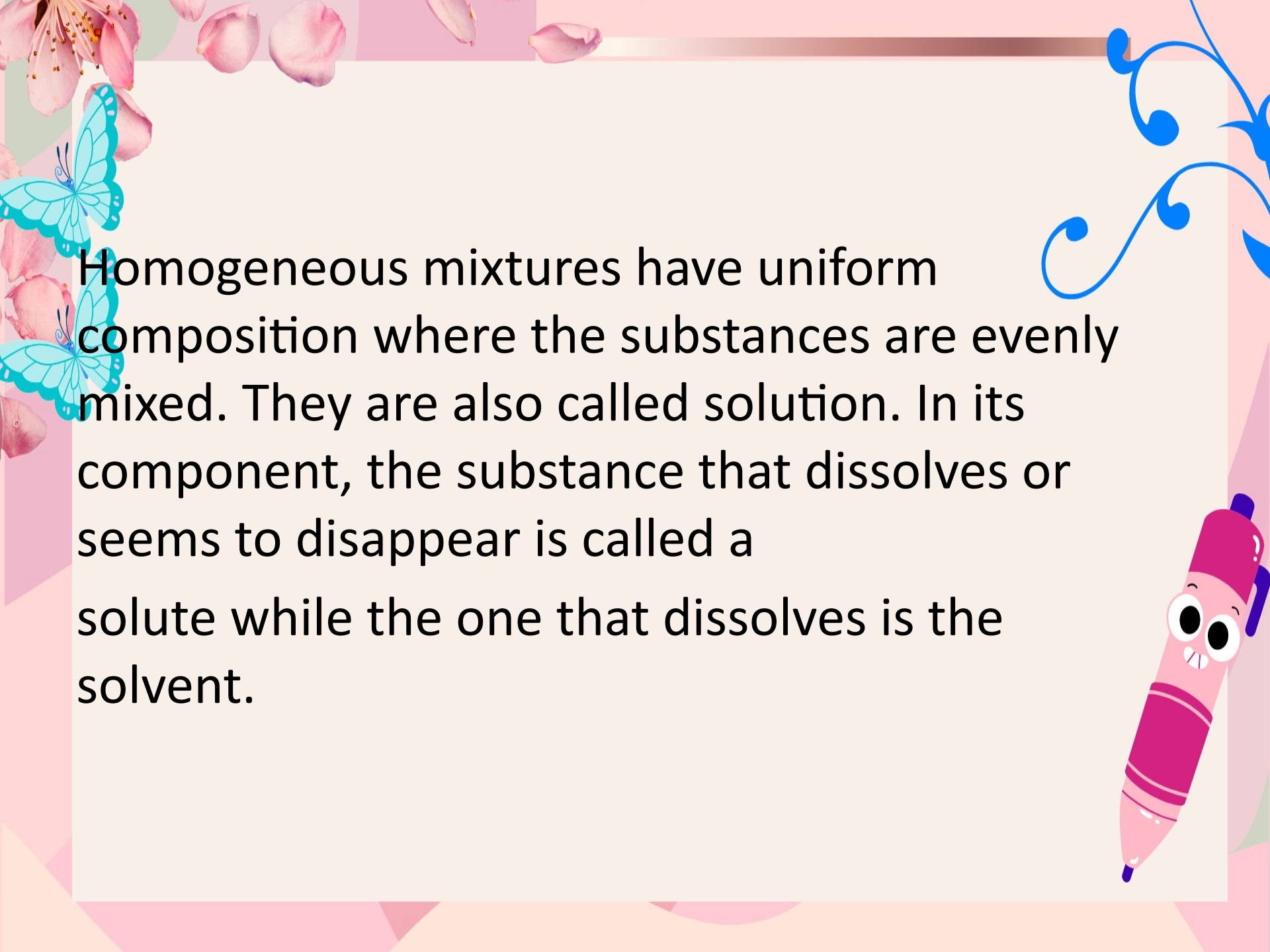
4. + =

5. + =

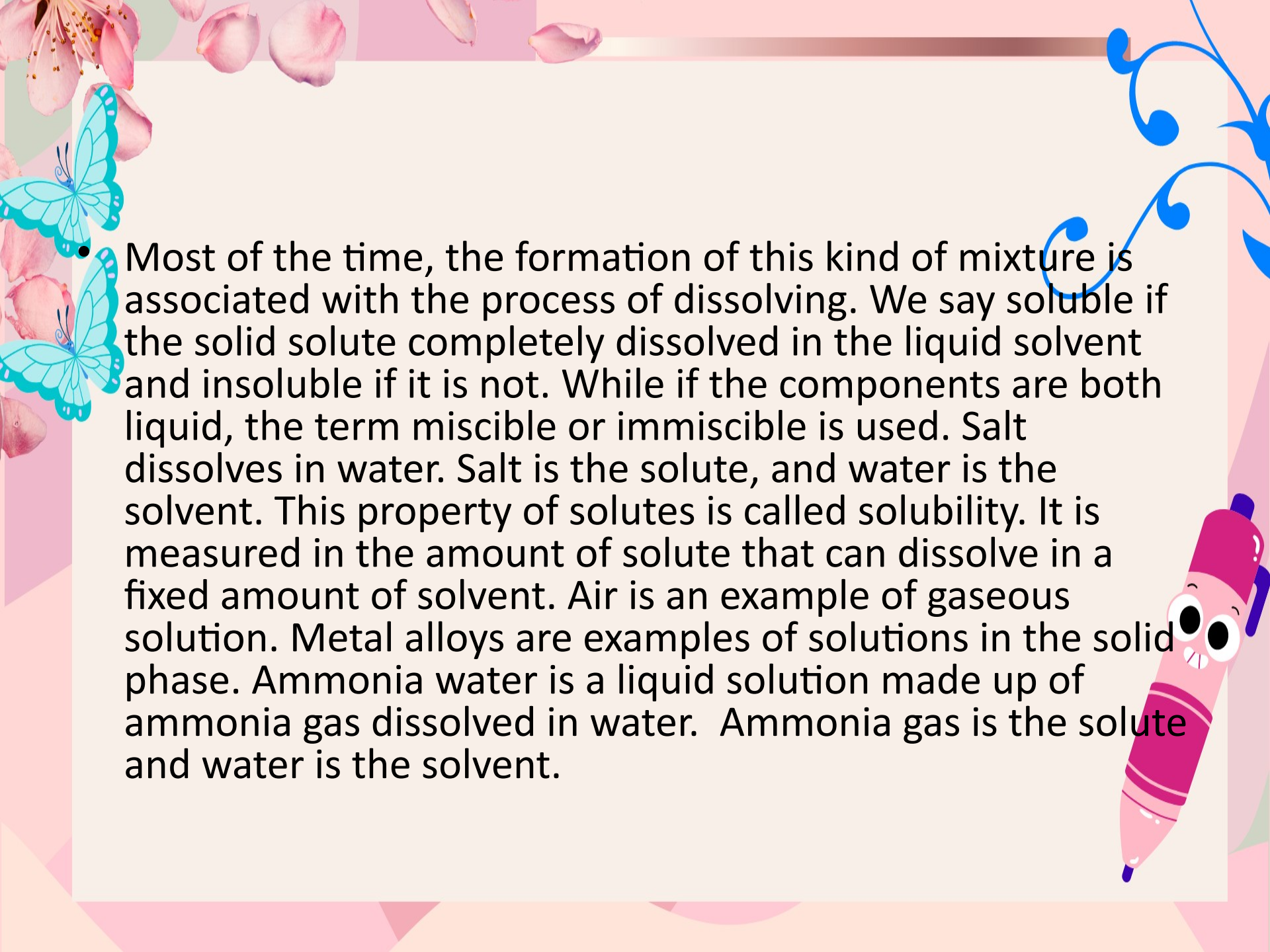


- Give another examples of solid homogeneous around you.

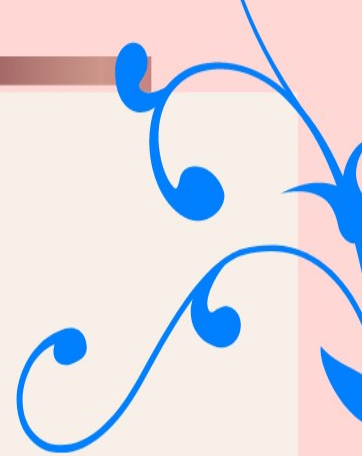


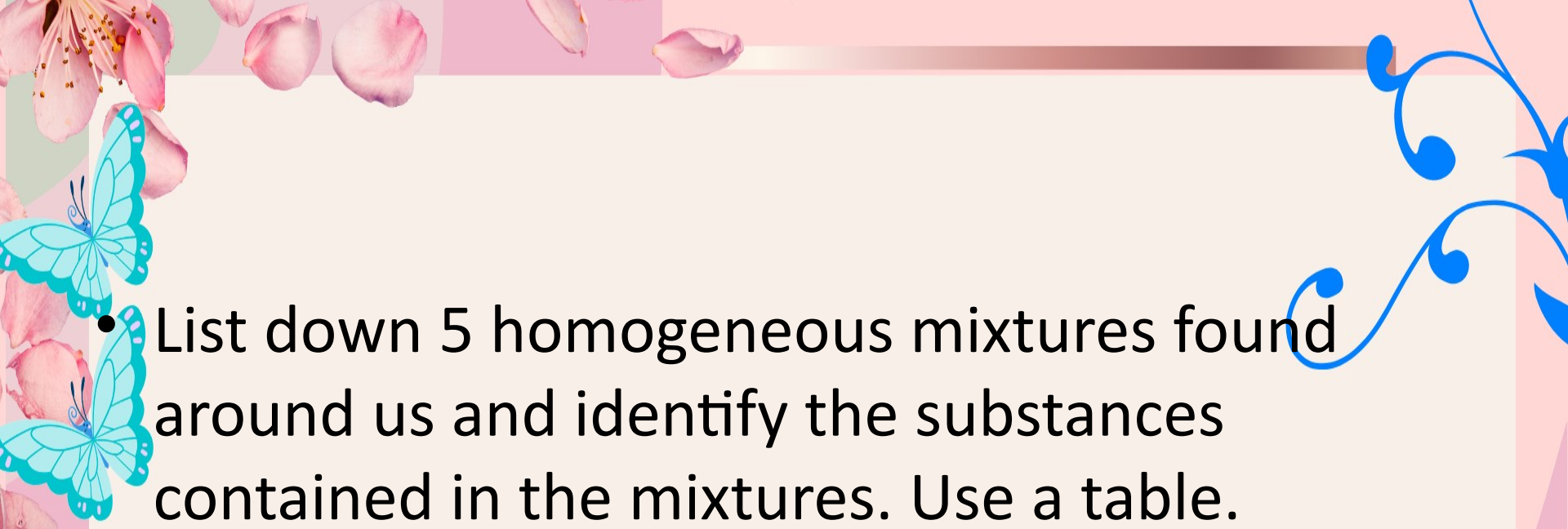


Homogeneous mixtures have uniform composition where the substances are evenly mixed. They are also called solution. In its component, the substance that dissolves or seems to disappear is called a solute while the one that dissolves is the solvent.

- 
- Most of the time, the formation of this kind of mixture is associated with the process of dissolving. We say soluble if the solid solute completely dissolved in the liquid solvent and insoluble if it is not. While if the components are both liquid, the term miscible or immiscible is used. Salt dissolves in water. Salt is the solute, and water is the solvent. This property of solutes is called solubility. It is measured in the amount of solute that can dissolve in a fixed amount of solvent. Air is an example of gaseous solution. Metal alloys are examples of solutions in the solid phase. Ammonia water is a liquid solution made up of ammonia gas dissolved in water. Ammonia gas is the solute and water is the solvent.

- **Discuss the concept of the lesson.**



- 
- List down 5 homogeneous mixtures found around us and identify the substances contained in the mixtures. Use a table.

Mixture	Substances



HOMOGENEOUS MIXTURES VERSUS HETEROGENEOUS MIXTURES

Homogenous mixtures have a uniform composition throughout the mixture

Components are not visible to the naked eye

The whole mixture is in the same phase

Particle size is often at atomic or molecular level

Components cannot be separated easily

Heterogenous mixtures have a mixed composition which may vary from point to point

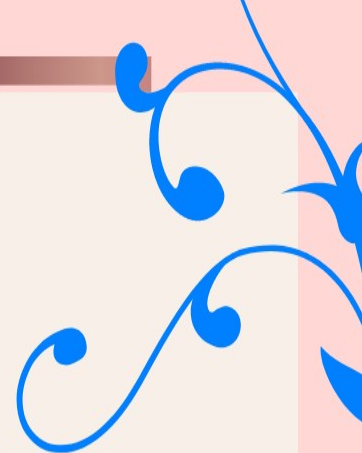
Components can be seen easily

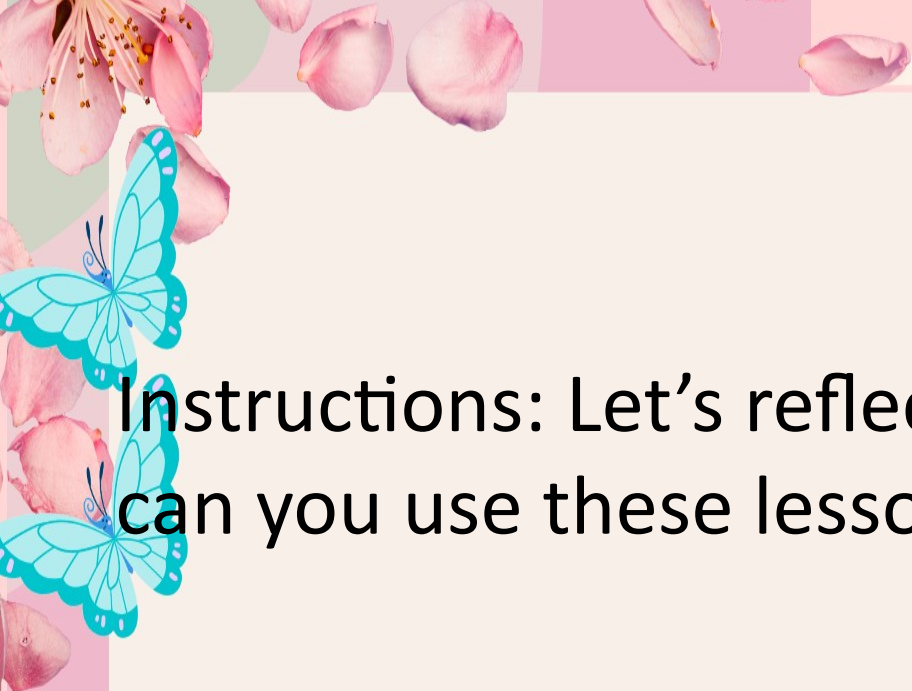
Substances can be of two phases and layers may separate

Heterogenous mixtures have large particle sizes

Components can be separated easily

Visit www.pediasa.com





Instructions: Let's reflect on our discussion. How can you use these lessons in real life situation?

